

Questions . One August 2022

(a) Difference between IN and OUT Instruction in Intel 8085

IN is used to receive data from an input device into the accumulator while OUT is used to send data from the accumulator to an output device

(b) Write an 8085 program to illustrate the difference between SHIFT and ROTATE

For ROTATE

MVI A, 45H

RLC

HLT

For SHIFT

MVI A, 45H

ADD A

HLT

(c) What are the control signals of DMA?
HOLD, HLDA

(d) Step-by-step procedure for implementing a subroutine in Intel 8085

1. Write the subroutine and give it a label
2. In the main program, call the subroutine using the CALL instruction
3. After execution the return address is pushed onto the stack
4. Write RET that pops the stored return address and returns execution to main program

(e) Describe the steps involved in an Opcode fetch operation in Intel 8085

1. The Program Counter contains the address of the next instruction
2. Address is placed on the address bus
3. CPU activates the Memory Read signal & Memory places the opcode onto the data bus
4. Opcode is loaded into the Instruction Register and the Program Counter is automatically incremented to point to the next instruction
5. The Control unit decodes the opcode and executes the instruction.

(f) Calculate the execution time of the code

i. Loop Count

LXI B, 2384H

DCX B, decrements the BC register pair until
 $BC = 0000H$

2384H to Decimal

$$\begin{aligned} &= (2 \times 4096) + (3 \times 256) + (8 \times 16) + 4 \\ &= 8192 + 768 + 128 + 4 \\ &= 9092 \end{aligned}$$

LXI B, 2384 H ≈ 10 T-States

Total T-States = 24

$$9092 - 1 = 9091 \times 24 = 218184$$

Final Iteration = $6 + 4 + 4 + 7 = 21$

$$= 10 + 218184 + 21 = 218215 \text{ T-States}$$